KAIZEN MATHS

Lesson starter

Straight Line Graphs and Gradients

Equation of Line

Name _____

S1

Find the equation of the line passing through (-3, 6) and (0, 2)

S2

Find the equation of the line passing through (0, 5) with gradient 2

S3

Find the equation of the line passing through (-2, 3) with y-intercept -3

S4

Find the equation of the line passing through (0, -4) and (2, 4)

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ALG-FCA4-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Equation of Line

Name

E1

Find the equation of the line passing through (-4, 0) with gradient -2

E2

Find the equation of the line passing through (-3, -5) with y-intercept 1

E3

Find the equation of the line passing through (-3, 6) and (4, -3)

ALG-FCA4-L3 · E

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Teacher answers & working

Straight Line Graphs and Gradients · Equation of Line

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S1

Find the equation of the line passing through (-3, 6) and (0, 2)

Answer:

$y = -\frac{4}{3}x + 2$

Working:

First find the gradient.

$m = \frac{2-6}{0--3} = \frac{-4}{3}$

Use y = mx + c with point (-3, 6).

$6 = \frac{-4}{3}(-3) + c$

$c = 2$

$y = -\frac{4}{3}x + 2$

S2

Find the equation of the line passing through (0, 5) with gradient 2

Answer:

$y = 2x + 5$

Working:

Use $y = mx + c$.

Substitute m = 2 and point (0, 5).

$5 = 2(0) + c$

$c = 5$

$y = 2x + 5$

S3

Find the equation of the line passing through (-2, 3) with y-intercept -3

Answer:

$y = -3x - 3$

Working:

The y-intercept gives the point (0, -3).

Find the gradient between (-2, 3) and (0, -3).

$m = \frac{3--3}{-2-0} = -3$

Use y = mx + c with c = -3.

$y = -3x - 3$

Teacher answers & working

Straight Line Graphs and Gradients · Equation of Line

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S4

Find the equation of the line passing through (0, -4) and (2, 4)

Answer:

$y = 4x - 4$

Working:

First find the gradient.

$m = \frac{4--4}{2-0} = 4$

Use y = mx + c with point (0, -4).

$-4 = 4(0) + c$

$c = -4$

$y = 4x - 4$

E1

Find the equation of the line passing through (-4, 0) with gradient -2

Answer:

$y = -2x - 8$

Working:

Use $y = mx + c$.

Substitute m = -2 and point (-4, 0).

$0 = -2(-4) + c$

$c = -8$

$y = -2x - 8$

E2

Find the equation of the line passing through (-3, -5) with y-intercept 1

Answer:

$y = 2x + 1$

Working:

The y-intercept gives the point (0, 1).

Find the gradient between (-3, -5) and (0, 1).

$m = \frac{-5-1}{-3-0} = 2$

Use y = mx + c with c = 1.

$y = 2x + 1$

Teacher answers & working

Straight Line Graphs and Gradients · Equation of Line

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E3

Find the equation of the line passing through (-3, 6) and (4, -3)

Answer:

$y = -\frac{9}{7}x + \frac{15}{7}$

Working:

First find the gradient.

$m = \frac{-3-6}{4--3} = \frac{-9}{7}$

Use y = mx + c with point (-3, 6).

$6 = \frac{-9}{7}(-3) + c$

$c = \frac{15}{7}$

$y = -\frac{9}{7}x + \frac{15}{7}$